step 1 : install required libraries

```
!pip install spacy pandas matplotlib seaborn emoji
!python -m spacy download en_core_web_sm
```

```
Collecting emoji
  Downloading emoji-2.15.0-py3-none-any.whl.metadata (5.7 kB)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.15)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.13)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.10)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.1.3)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.5.2)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.10)
Requirement already satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.3)
Requirement already satisfied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.21.1)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (4.67.1)
Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.2)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.32.4)
Requirement already satisfied: pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.12.3)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.1.6)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from spacy) (75.2.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (25.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.61.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.4.9)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.1)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4->spacy) (0.7.0)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4->spacy) (2.41.4)
Requirement already satisfied: typing-extensions>=4.14.1 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4->spacy) (4.15.0)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic!=1.8,!=1.8.1,<3.0.0,>=1.7.4->spacy) (0.4.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (2026.1.4)
Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->spacy) (1.3.3)
Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.12/dist-packages (from thinc<8.4.0,>=8.3.4->spacy) (0.1.5)
Requirement already satisfied: click>=8.0.0 in /usr/local/lib/python3.12/dist-packages (from typer-slim<1.0.0,>=0.3.0->spacy) (8.3.1)
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2->spacy) (0.23.0)
Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python3.12/dist-packages (from weasel<0.5.0,>=0.4.2->spacy) (7.5.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->spacy) (3.0.3)
Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart-open<8.0.0,>=5.2.1->weasel<0.5.0,>=0.4.2->spacy) (2.0.1)
Downloading emoji-2.15.0-py3-none-any.whl (608 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 608.4/608.4 kB 7.5 MB/s eta 0:00:00
Installing collected packages: emoji
Successfully installed emoji-2.15.0
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any.whl (12.8 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 74.0 MB/s eta 0:00:00
✔ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

step 2 : import libraries

```
import pandas as pd
import re
import spacy
import emoji
import matplotlib.pyplot as plt
from collections import Counter
import seaborn as sns
```

step 3: load dataset

```
from google.colab import files

# This will open a file picker dialog to let you upload the file
uploaded = files.upload()

for fn in uploaded.keys():
  print(f'User uploaded file "{fn}" with length {len(uploaded[fn])} bytes')
```

Choose files Tweets.csv

Tweets.csv(text/csv) - 3421431 bytes, last modified: 22/01/2026 - 100% done

```
Saving Tweets.csv to Tweets.csv
User uploaded file "Tweets.csv" with length 3421431 bytes
```

```
df = pd.read_csv('Tweets.csv')
display(df.head())
```

	tweet_id	airline_sentiment	airline_sentiment_confidence	negativereason	negativereason_confidence	airline	airline_sentiment_gold	name	ne
0	570306133677760513	neutral	1.0000	NaN	NaN	Virgin America	NaN	cairdin	
1	570301130888122368	positive	0.3486	NaN	0.0000	Virgin America	NaN	jnardino	
2	570301083672813571	neutral	0.6837	NaN	NaN	Virgin America	NaN	yvonnalynn	
3	570301031407624196	negative	1.0000	Bad Flight	0.7033	Virgin America	NaN	jnardino	
4	570300817074462722	negative	1.0000	Can't Tell	1.0000	Virgin America	NaN	jnardino	

step 4: preprocess tweets

```
def clean_tweet(text):
    text = str(text)
    text = re.sub(r"http\S+|www\S+|https\S+", "", text)  # remove URLs
    text = re.sub(r"@\w+", "", text)  # remove mentions
    text = emoji.replace_emoji(text, replace="")  # remove emojis
    text = re.sub(r"#", "", text)  # keep hashtag word, remove #
    text = re.sub(r"\s+", " ", text).strip()  # remove extra spaces
    return text

# Apply cleaning
df['clean_text'] = df['text'].apply(clean_tweet)

# Check cleaned text
df[['text', 'clean_text']].head()
```

	text	clean_text
0	@VirginAmerica What @dhepburn said.	What said.
1	@VirginAmerica plus you've added commercials t...	plus you've added commercials to the experienc...
2	@VirginAmerica I didn't today... Must mean I n...	I didn't today... Must mean I need to take ano...
3	@VirginAmerica it's really aggressive to blast...	it's really aggressive to blast obnoxious "ent...
4	@VirginAmerica and it's a really big bad thing...	and it's a really big bad thing about it

```
Start coding or generate with AI.
```

step 5 : load spacy model

```
nlp = spacy.load("en_core_web_sm")
```

step 6 : process cleaned text using spacy

```
docs = list(nlp.pipe(df['clean_text']))
```

step 7: extract lemmas

```
df['lemmas'] = [
    [token.lemma_ for token in doc if not token.is_stop]
    for doc in docs
]
```

step 8 : Extract POS tags

```
df['pos_tags'] = [
    [token.pos_ for token in doc]
    for doc in docs
]

df.head()
```

	tweet_id	airline_sentiment	airline_sentiment_confidence	negativereason	negativereason_confidence	airline	airline_sentiment_gold	name	ne
0	570306133677760513	neutral	1.0000	NaN	NaN	Virgin America	NaN	cairdin	
1	570301130888122368	positive	0.3486	NaN	0.0000	Virgin America	NaN	jnardino	
2	570301083672813571	neutral	0.6837	NaN	NaN	Virgin America	NaN	yvonnalynn	
3	570301031407624196	negative	1.0000	Bad Flight	0.7033	Virgin America	NaN	jnardino	
4	570300817074462722	negative	1.0000	Can't Tell	1.0000	Virgin America	NaN	jnardino	

Next steps: Generate code with df New interactive sheet

step 9 :create pipeline component

```
from spacy.language import Language

@Language.component("hashtag_detector")
def hashtag_detector(doc):
    doc._.hashtags = [token.text for token in doc if token.text.startswith("#")]
    return doc
```

```
from spacy.tokens import Doc

Doc.set_extension("hashtags", default=[], force=True)
```

step 10 :Add component to spacy pipeline

```
nlp.add_pipe("hashtag_detector", last=True)
```

```
hashtag_detector
def hashtag_detector(doc)

<no docstring>
```

step 11 : apply updated pipeline

```
docs = list(nlp.pipe(df['clean_text']))
df['hashtags'] = [doc._.hashtags for doc in docs]

df[['clean_text', 'hashtags']].head()
```

	clean_text	hashtags
0	What said.	[]
1	plus you've added commercials to the experienc...	[]
2	I didn't today... Must mean I need to take ano...	[]
3	it's really aggressive to blast obnoxious "ent...	[]
4	and it's a really big bad thing about it	[]

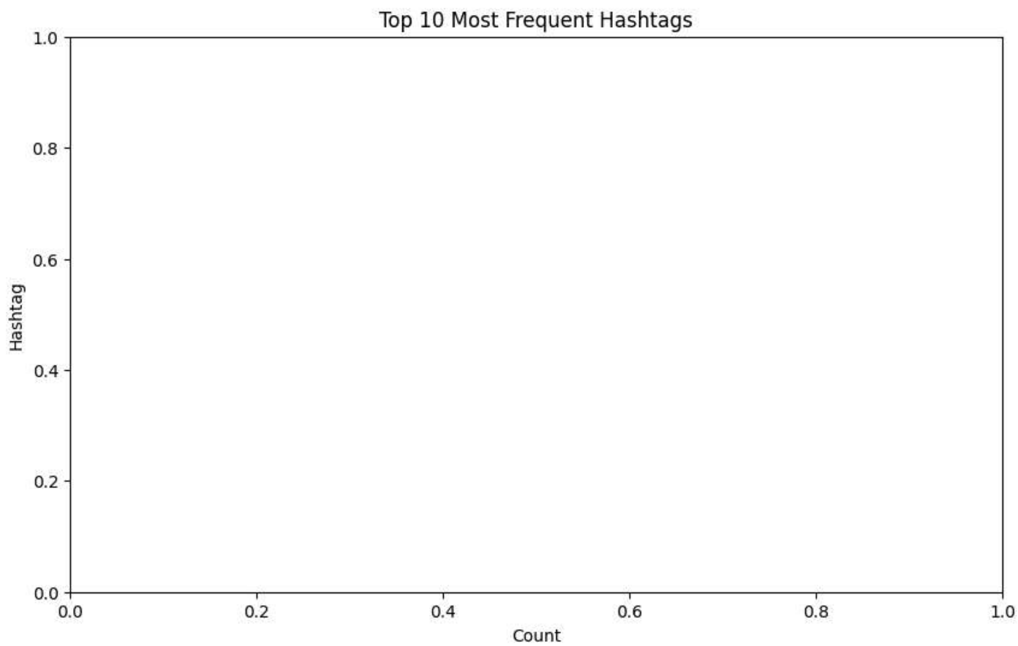
step 12:visalize frequent hashtags

```
all_hashtags = [hashtag for sublist in df['hashtags'] for hashtag in sublist]
hashtag_counts = Counter(all_hashtags)

# Get the 10 most common hashtags
most_common_hashtags = hashtag_counts.most_common(10)

# Prepare data for plotting
hashtags_df = pd.DataFrame(most_common_hashtags, columns=['Hashtag', 'Count'])

# Plotting
plt.figure(figsize=(10, 6))
sns.barplot(x='Count', y='Hashtag', data=hashtags_df, palette='viridis')
plt.title('Top 10 Most Frequent Hashtags')
plt.xlabel('Count')
plt.ylabel('Hashtag')
plt.show()
```



step 13:POS distribution in negative tweets

```
negative_df = df[df['airline_sentiment'] == 'negative']

neg_docs = list(nlp.pipe(negative_df['clean_text']))
neg_pos = [token.pos_ for doc in neg_docs for token in doc]

pos_freq = Counter(neg_pos)

plt.figure()
plt.bar(pos_freq.keys(), pos_freq.values())
plt.title("POS Distribution in Negative Tweets")
plt.show()
```